

- Q.28 Explain the working principle of inclined tube manometers.
- Q.29 Explain with neat sketch the working principle of elastic pressure transducers used for measuring pressure.
- Q.30 Describe with application closed loop control system.
- Q.31 Define input-step, Ramp, sinusoidal and pulse.
- Q.32 Define with neat and clean sketch strip chart recorder.
- Q.33 Explain valve actuator and valve positioning.
- Q.34 Explain with neat sketch radiation level detector.
- Q.35 Define controlled variable manipulated variable and disturbances.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain with block diagram representation the concept and components of automatic control system.
- Q.37 Describe with neat diagram the working principle and constructional details of a radiation pyrometer.
- Q.38 What is thermistor? Explain with a neat sketch the working and constructional details of a thermistor.

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Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Set point of a system is also called
- Manipulated variable
 - Desired variable
 - Controlled variable
 - Disturbance
- Q.2 Span is
- Low & high value
 - Low value
 - Difference between high & low value
 - High value
- Q.3 Cascade control system employs
- Two feed forward
 - Two feed back
 - One feed forward
 - One feed back
- Q.4 Which system has tendency to oscillate.
- Open loop system
 - Closed loop system
 - Ratio controller
 - Cascade control

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- Q.5 Thermopile measures the
- Temperature
 - EMFC
 - Current
 - Flow rate
- Q.6 Optical level detector uses
- Sound
 - Gamma rays
 - Light
 - None
- Q.7 A process control system consists of
- 10 elements
 - 6 elements
 - 2 elements
 - 4 elements
- Q.8 The performance of a capacitance level indicator is severely affected by dirt because it changes the
- Area of plate
 - Distance between two plates
 - Dielectric constant
 - None of the above
- Q.9 The smallest change in the value of a measured variable to which instrument responds is called
- Static error
 - Sensitivity
 - Reproducibility
 - Dead zone
- Q.10 Working principle of radiation pyrometer is based on the
- Wien's law
 - Stefan-Boltz man law
 - Kirchoff's law
 - See beck effect

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define precision.
- Q.12 Define sensitivity.
- Q.13 Define transfer function.
- Q.14 Define drift.
- Q.15 Write the main function of control valve.
- Q.16 Define calibration.
- Q.17 What are the main functions of pyrameters?
- Q.18 Name any two analyzers.
- Q.19 Define PH.
- Q.20 Define dead zone.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Convert 100^oc into kelvin, ^oF and ^oRankine.
- Q.22 Write different static errors.
- Q.23 Explain with neat sketch sight glass liquid level measuring instrument.
- Q.24 Explain working of PH meter in brief.
- Q.25 Explain with block diagram the functional elements of an instrument.
- Q.26 Explain oxygen analyzer.
- Q.27 Explain with neat sketch the working of bimetallic thermometer.